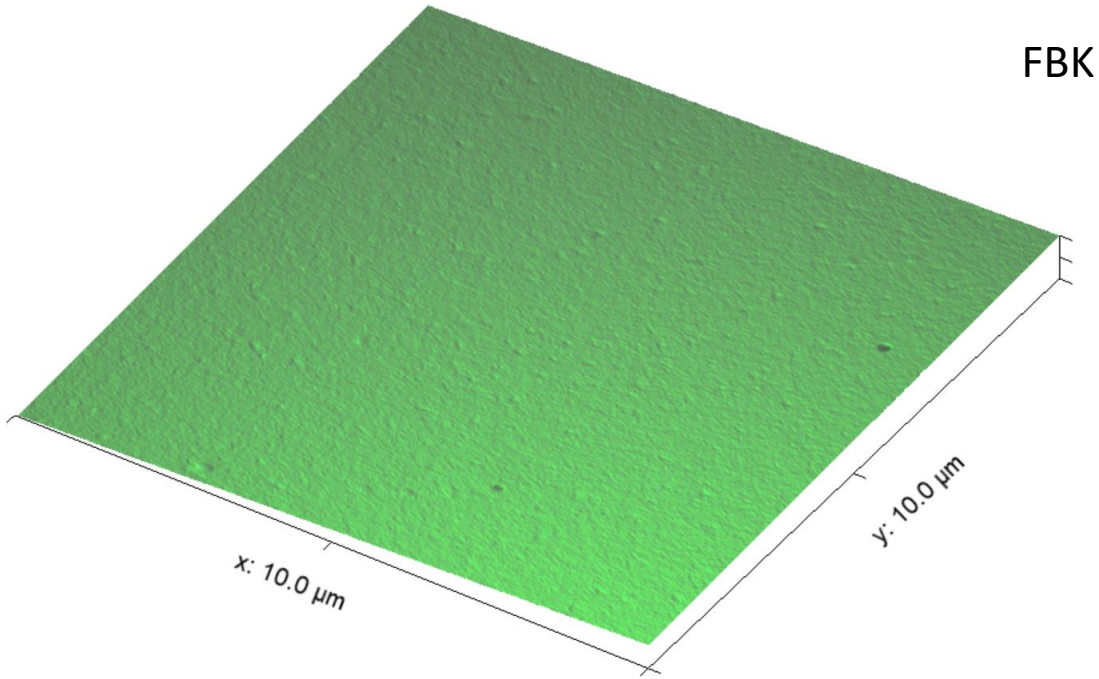
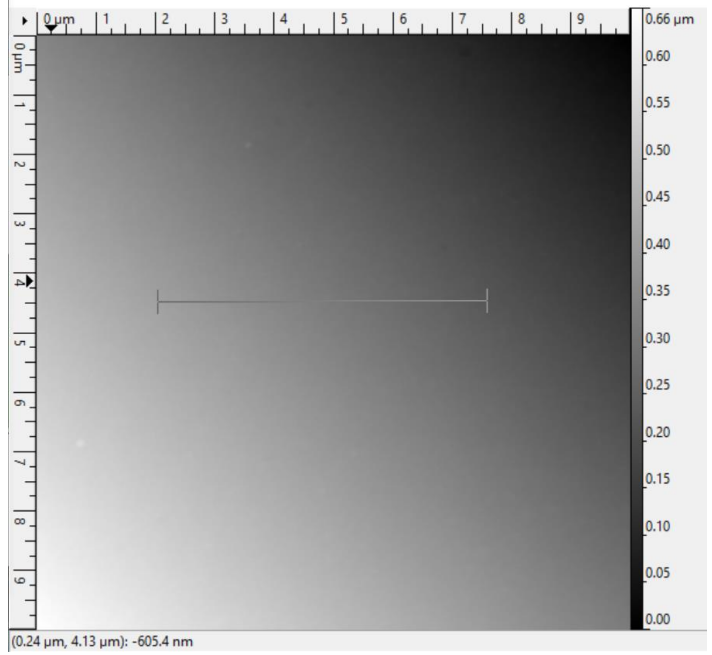
Study of Topography of FBK and VUV using Atomic Force Microscopy (AFM)

Prabandha Nakarmi

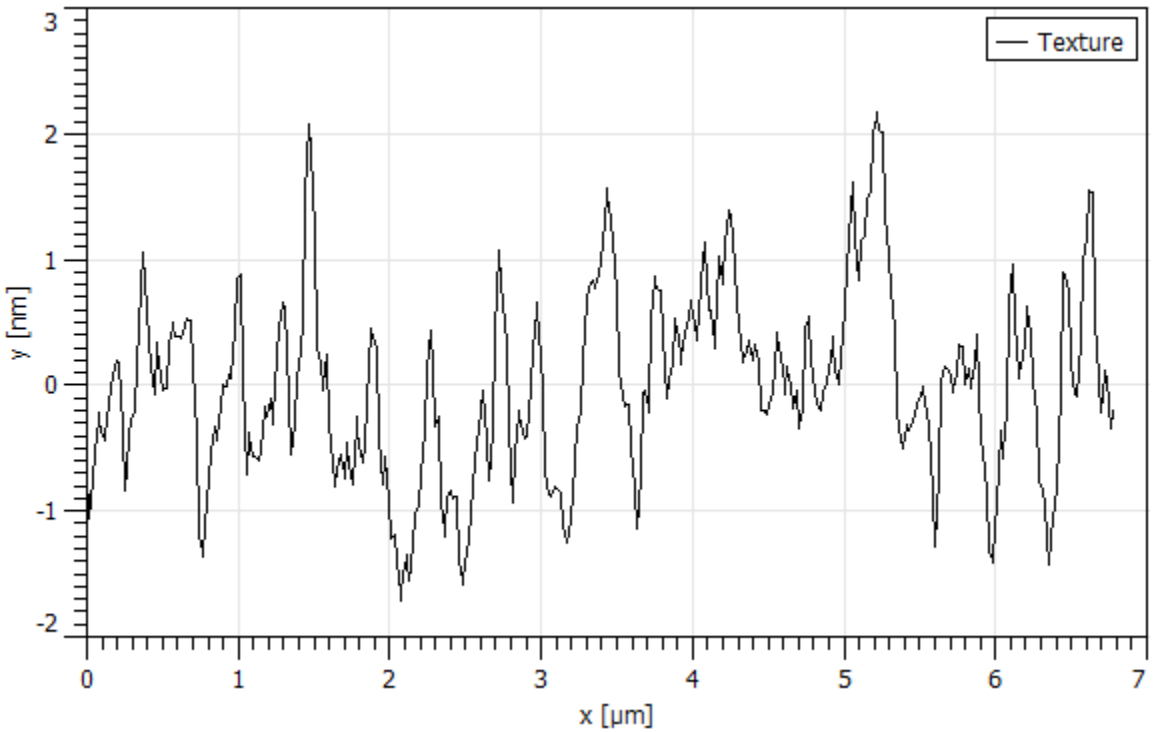
The University of Alabama

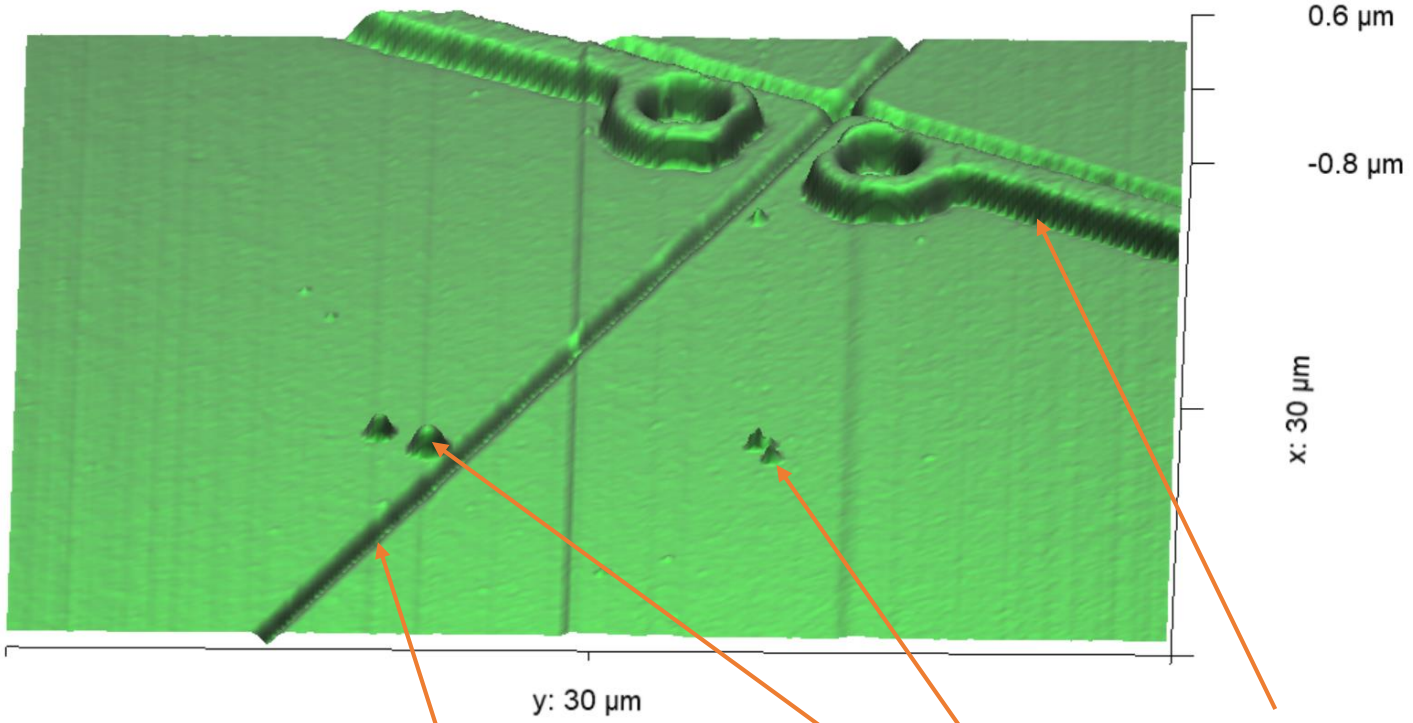
FBK Surface Topography





The plain surface at the center of the active region of FBK has ~ 3.5 nm of roughness.

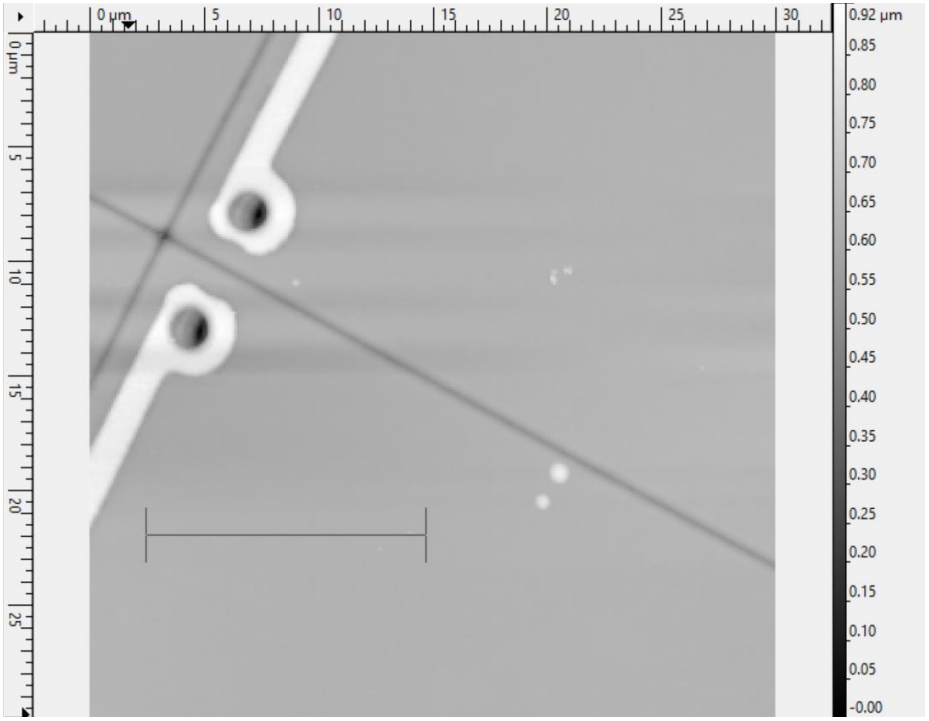


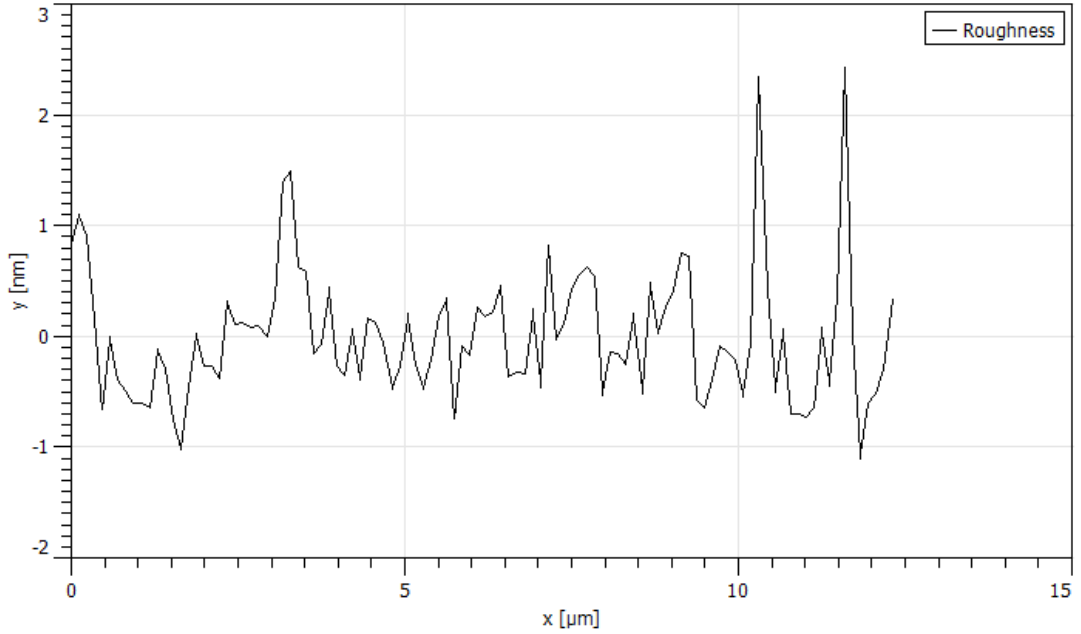


Polysilicon resistor

Dust

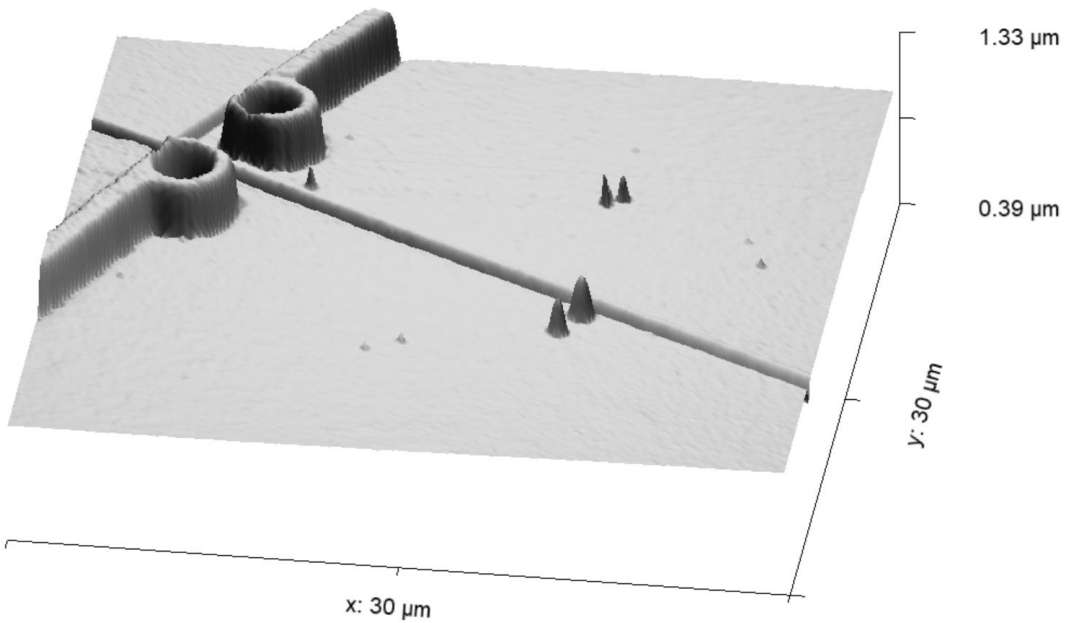
Connection of two cells

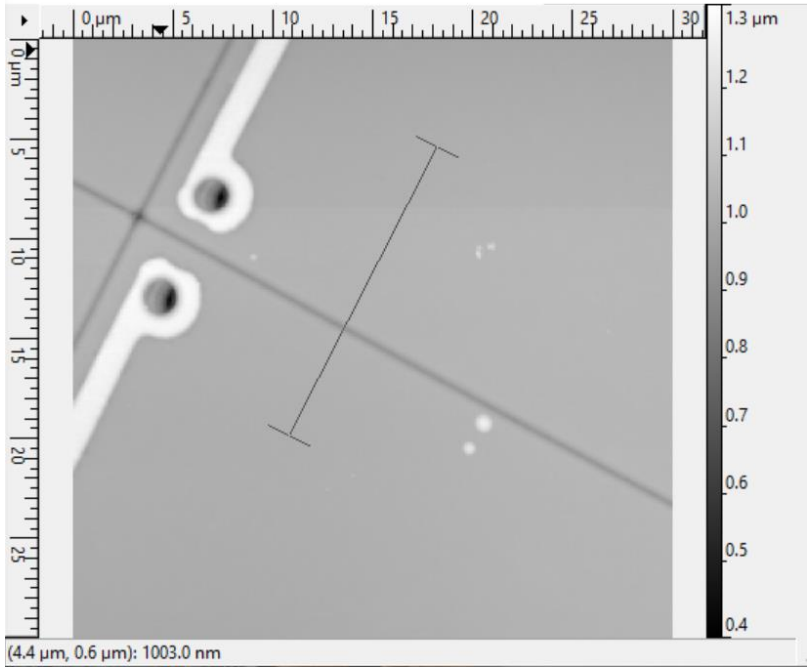




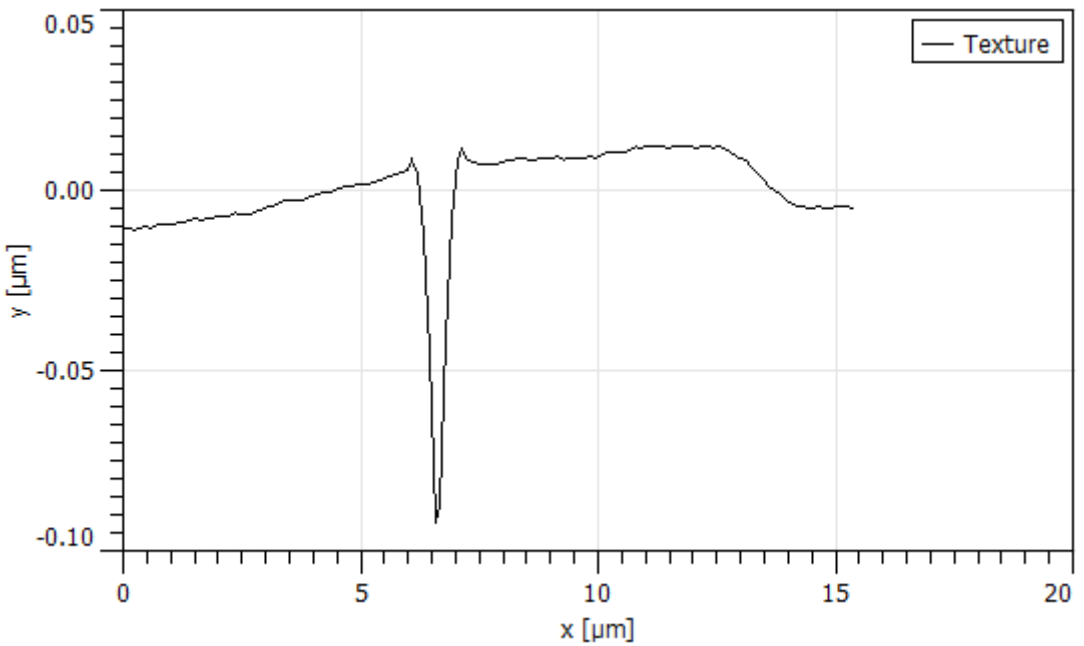
The texture of the plain surface of FBK is found to be ~3.5 nm scale which is same as at the center of active cell.

FBK Surface Topography

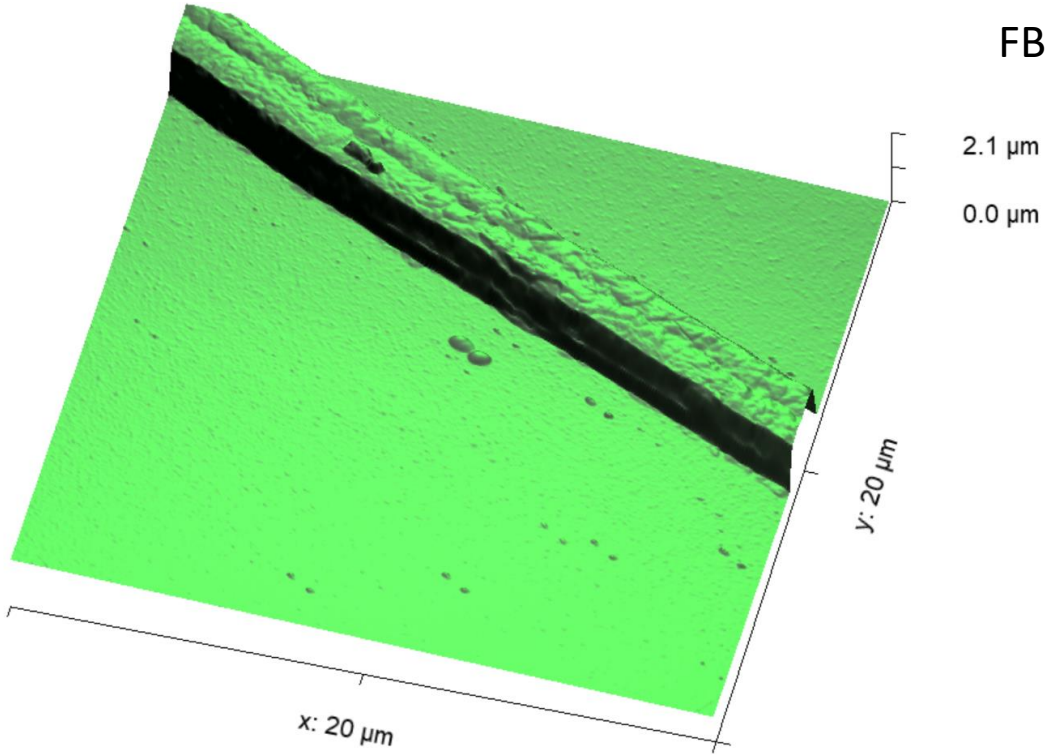


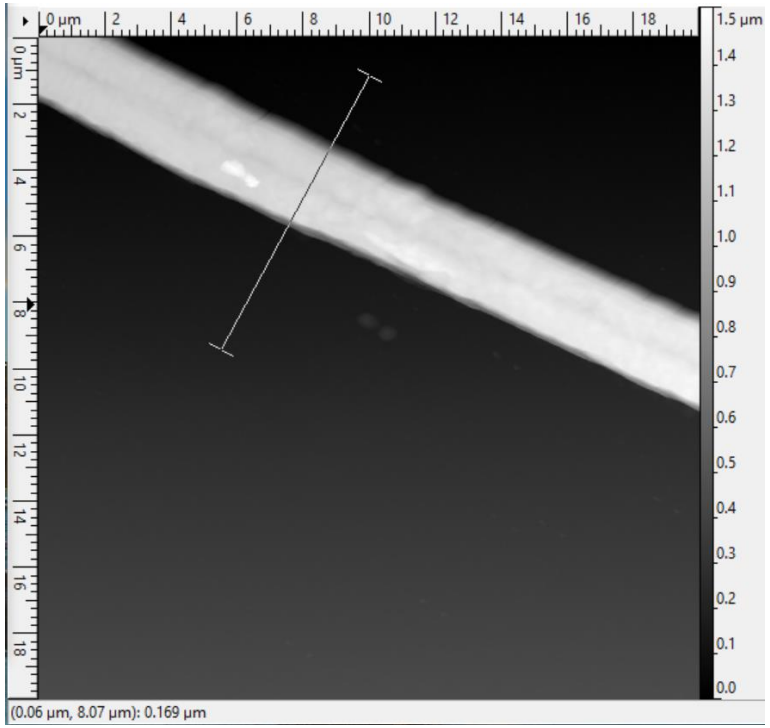


The connection of two cells in FBK is found to be ~90 nm deep with ~20 nm of height above the surface.

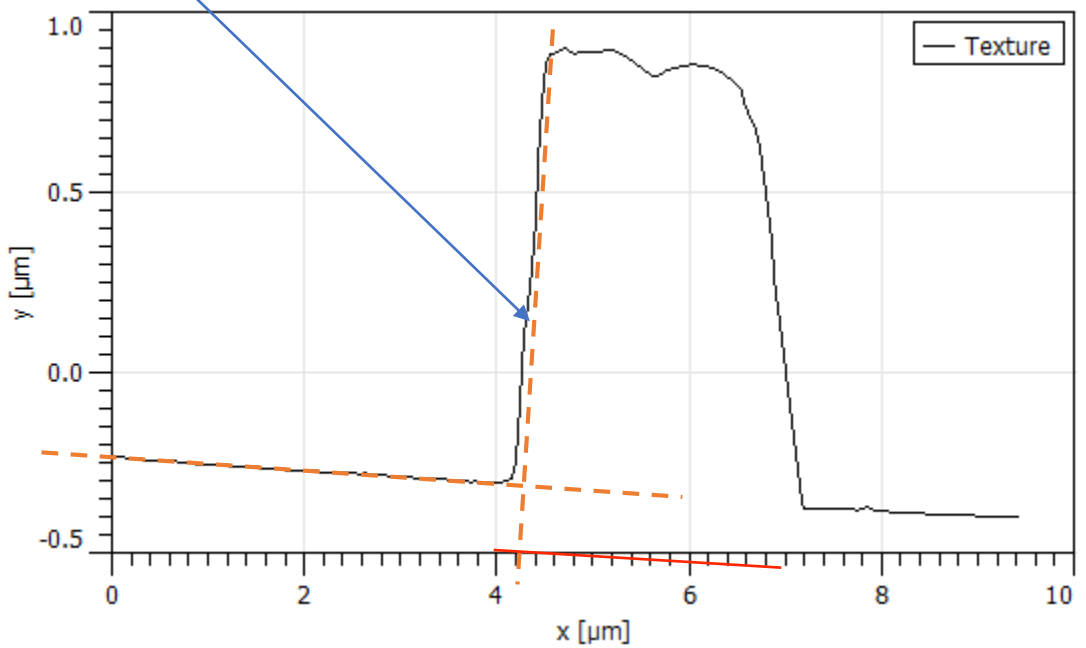


FBK Surface Topography



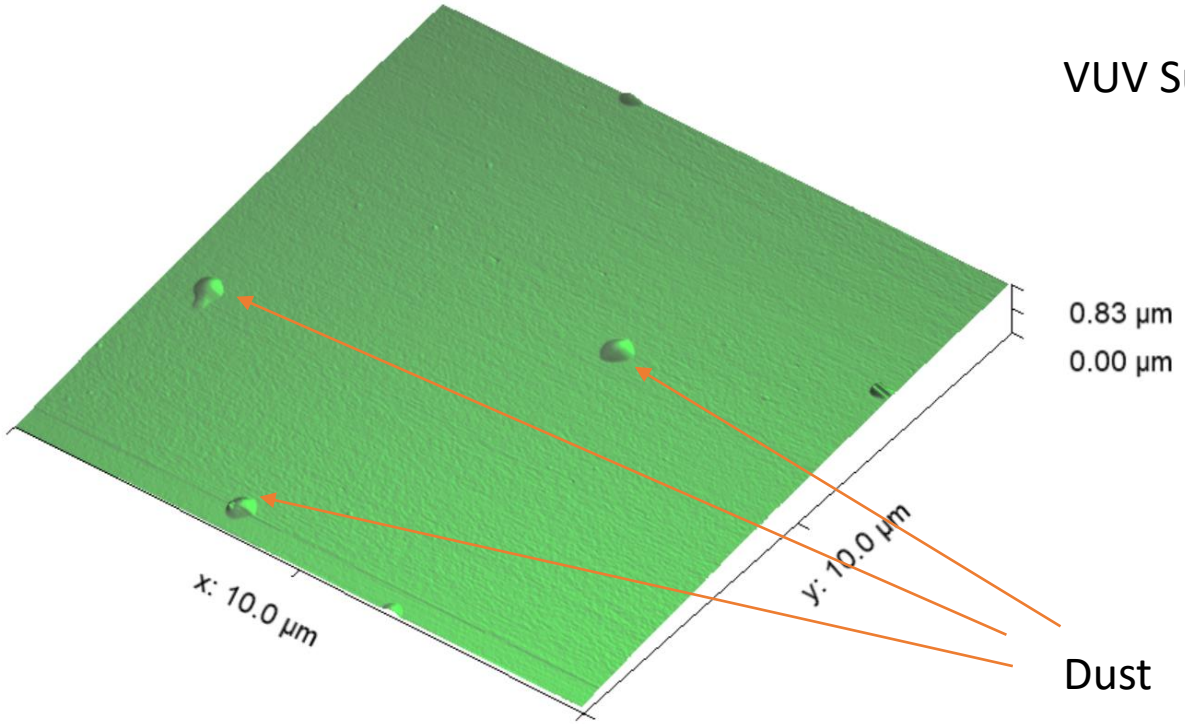


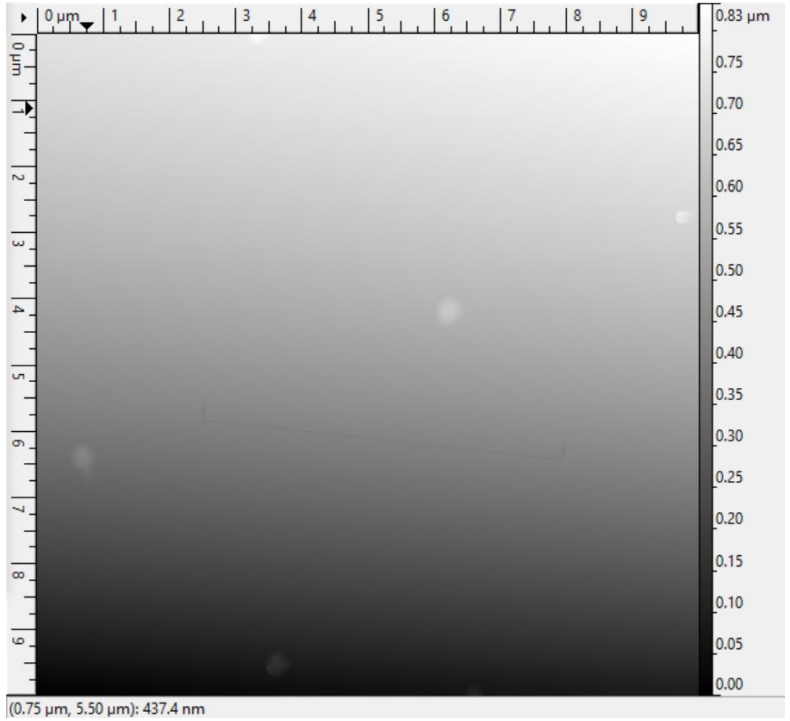
Angle (β)



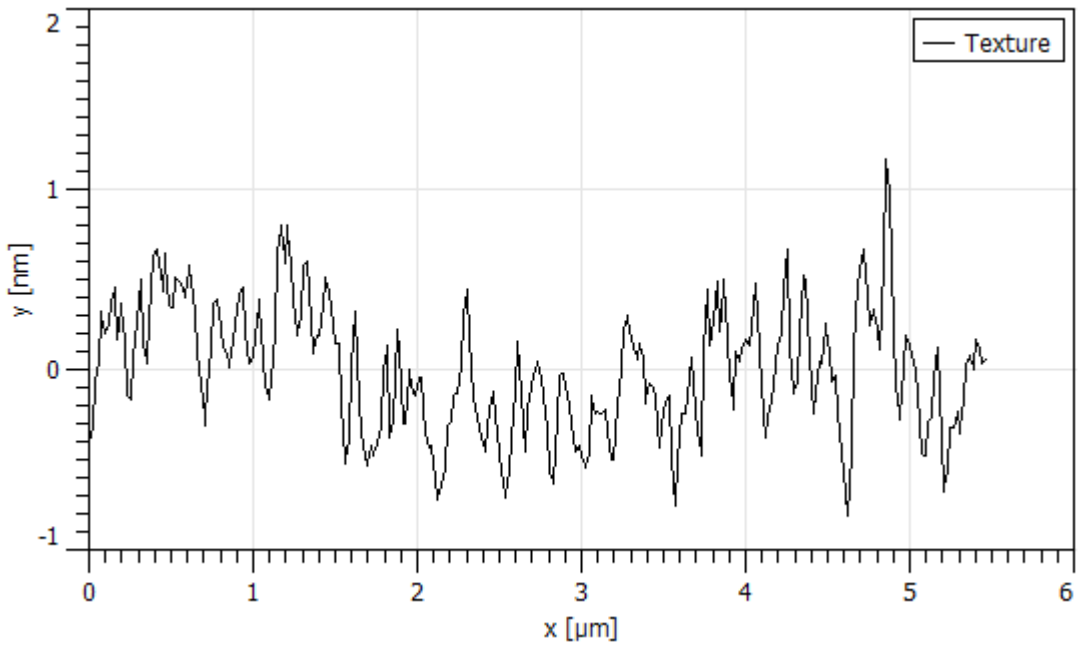
- The polysilicon resistor between the two cells is found to be ~ 1.3 um.
- The angle (β) can be calculated and is found to be ~ 3 degrees.

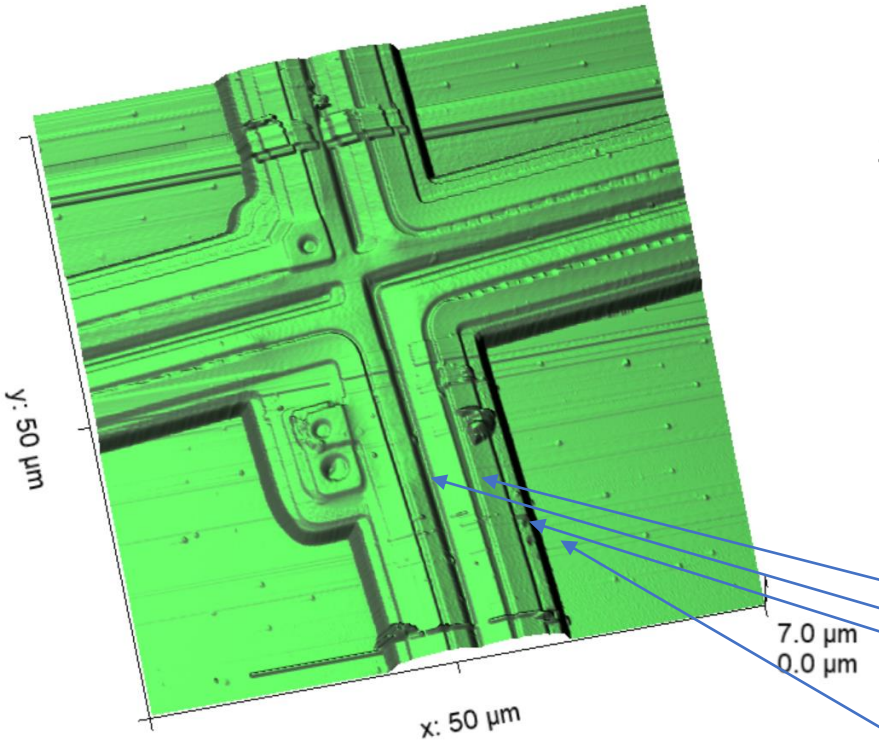
VUV Surface Topography



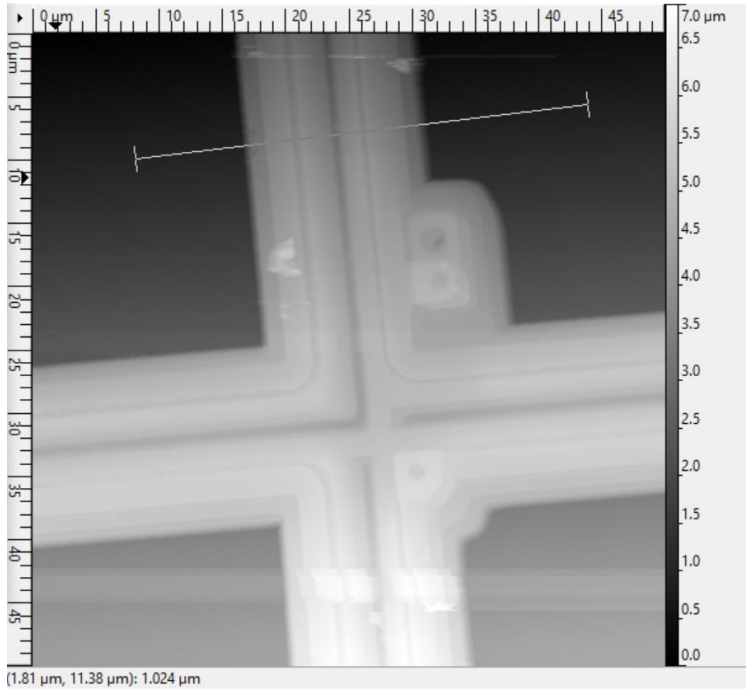


- Roughness in the plain surface for VUV is found to be ~ 2 nm.

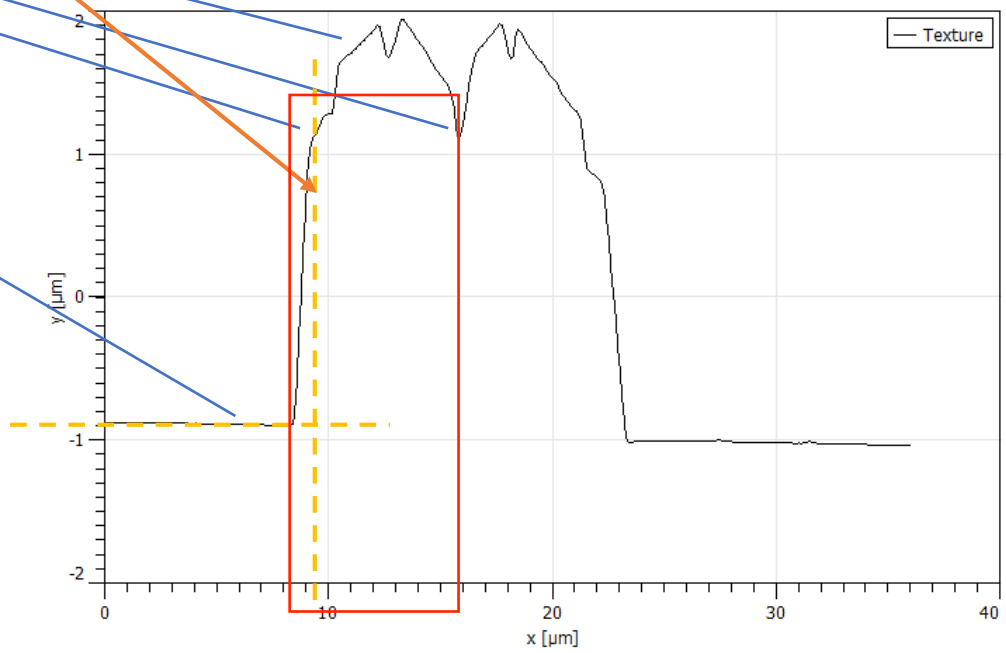




Surface Topography of VUV



Angle (β)



- The surface texture in the connections of two active area is 2 um – 3 um. The few peaks and valleys at the top can be compared to the 3D image to the top left.
- The angle (β) can be calculated from the texture profile and is found to be ~25 degrees